

**Project Report:**

# **SOC-Based Network Attack Detection and Traffic Analysis using Wireshark**

**Prepared By:**

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## **Introduction**

Cybersecurity has become one of the most critical domains in modern information technology due to the increasing number of cyberattacks targeting organizations, institutions, and individuals. Modern enterprises rely heavily on computer networks for communication, data transfer, remote administration, cloud services, and business operations. As networks continue to expand, attackers continuously attempt to exploit vulnerabilities within these infrastructures.

A Security Operations Center (SOC) is responsible for continuously monitoring network traffic, detecting malicious activities, analyzing threats, and responding to security incidents. SOC analysts play an important role in identifying suspicious behavior within networks before attackers achieve unauthorized access or data theft.

Network attacks generally follow multiple stages. Attackers initially perform reconnaissance to identify active hosts and exposed services. Once information is collected, attackers attempt service enumeration, vulnerability identification, password attacks, exploitation, and privilege escalation. Detecting these early-stage activities is extremely important because they often indicate an ongoing intrusion attempt.

Packet analysis is one of the most effective methods used in SOC environments for threat detection. By inspecting packets traveling across the network, analysts can identify suspicious patterns such as:

- ICMP reconnaissance
- Port scanning activities
- Unauthorized service enumeration
- Repeated authentication attempts
- Malicious TCP communication
- Data exfiltration attempts
- Protocol anomalies

Wireshark is one of the most widely used network protocol analyzers in cybersecurity and digital forensics. It allows analysts to capture and inspect packets in real time. Wireshark supports deep inspection of hundreds of protocols and provides detailed visibility into TCP/IP communication.

In this project, a virtualized SOC environment was created using VMware Workstation. Kali Linux was used as the attacker machine, while Windows 10 acted as the victim machine. Multiple attack simulations were performed including:

- ICMP reconnaissance
- TCP SYN scanning
- Service version enumeration
- HTTP traffic analysis
- Remote Desktop Protocol (RDP) password attack simulation

The generated traffic was captured and analyzed using Wireshark. Different packet types, TCP flags, protocols, and communication behaviors were examined to understand how SOC analysts investigate suspicious network activities.

**This project provides practical exposure to:**

- Network traffic monitoring
- Attack detection methodologies
- Packet-level investigation
- Reconnaissance analysis
- Authentication attack detection
- SOC investigation workflow
- Incident analysis techniques

The primary purpose of this project is to demonstrate how packet analysis can be used for identifying malicious activities in enterprise networks and how SOC analysts utilize network evidence to investigate cyber incidents.

## **Objectives**

**The main objectives of this project are:**

- To create a virtual SOC-based network attack simulation environment
- To analyze network packets using Wireshark
- To detect reconnaissance activities using ICMP analysis
- To identify SYN scan behavior during port scanning
- To analyze service enumeration traffic
- To detect suspicious RDP authentication attempts
- To study TCP packet behavior during attacks
- To understand the role of packet analysis in SOC operations

## **Tools and Technologies Used**

| <b>Tool/Technology</b> | <b>Purpose</b>                      |
|------------------------|-------------------------------------|
| Kali Linux             | Attacker machine                    |
| Windows 10             | Victim machine                      |
| Wireshark              | Network traffic analysis            |
| Nmap                   | Network reconnaissance and scanning |
| Hydra                  | Password attack simulation          |
| VMware Workstation     | Virtualized lab environment         |
| TCP/IP Protocol Suite  | Communication protocols             |

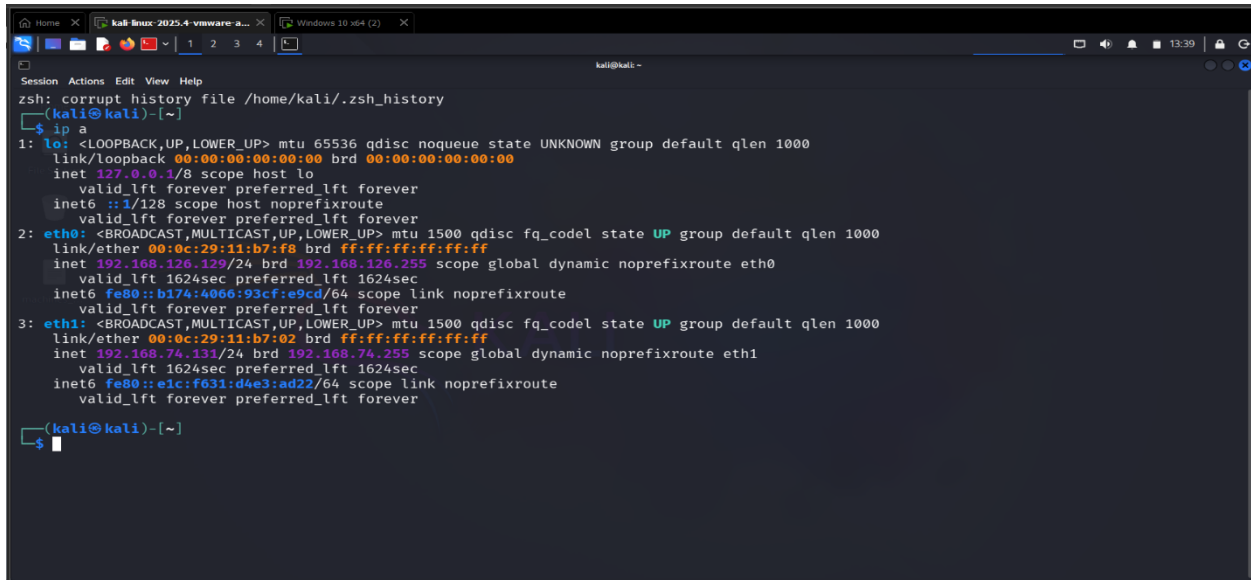
## Lab Environment Setup

The project was implemented inside VMware Workstation using an isolated virtual network environment.

### System Configuration

| Machine    | Role     | IP Address     |
|------------|----------|----------------|
| Kali Linux | Attacker | 192.168.74.131 |
| Windows 10 | Victim   | 192.168.74.130 |

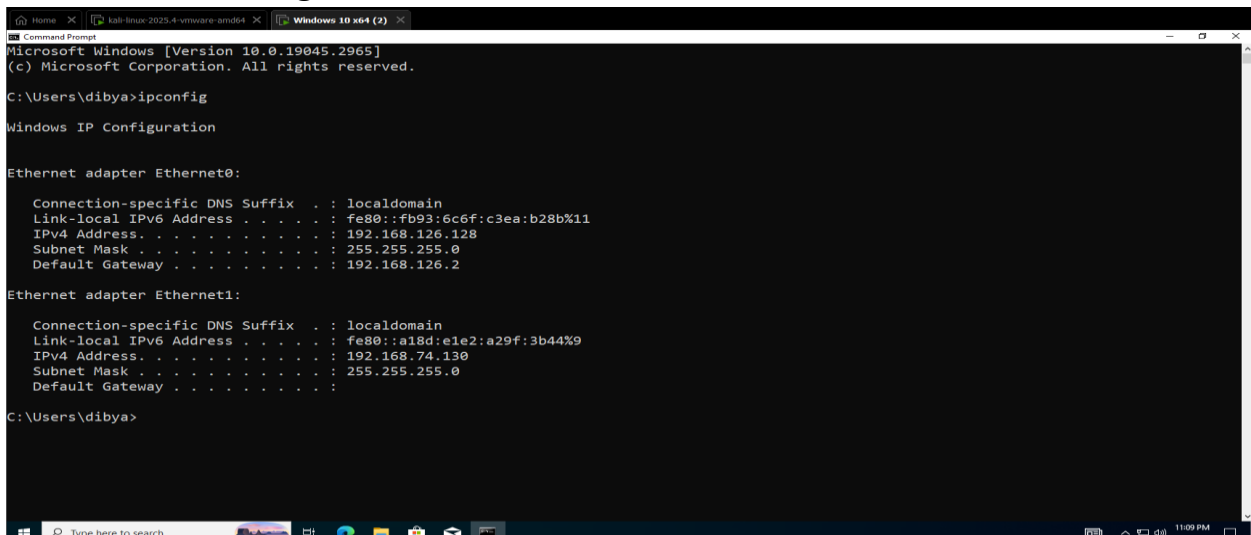
### Kali Linux IP Configuration



```
kali@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:11:b7:f8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.126.129/24 brd 192.168.126.255 scope global dynamic noprefixroute eth0
        valid_lft 1624sec preferred_lft 1624sec
    inet6 fe80::b174:4066:93cf:e9cd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:11:b7:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.74.131/24 brd 192.168.74.255 scope global dynamic noprefixroute eth1
        valid_lft 1624sec preferred_lft 1624sec
    inet6 fe80::e1c:f631:d4e3:ad22/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

The above figure shows the IP configuration of the Kali Linux attacker machine.

### Windows 10 IP Configuration



```
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dibya>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::fb93:6c6f:c3ea:b28b%11
    IPv4 Address. . . . . : 192.168.126.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.126.2

Ethernet adapter Ethernet1:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::a18d:e1e2:a29f:3b44%9
    IPv4 Address. . . . . : 192.168.74.130
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

C:\Users\dibya>
```

The above figure shows the IP configuration of the Windows 10 victim machine.

## **Attack Simulation and Detection**

Cyber attackers typically follow a structured methodology while targeting systems within a network. Understanding attacker behavior is essential for SOC analysts because each attack stage generates identifiable network patterns.

In this project, several attack simulations were performed in a controlled virtual environment.

The attacks were intentionally selected to represent common activities observed during the early stages of cyber intrusions.

The attack workflow followed this sequence:

1. Reconnaissance
2. Host discovery
3. Port scanning
4. Service enumeration
5. Authentication attack attempts
6. Traffic analysis and detection

Each stage generated different network signatures that were analyzed using Wireshark.

### **5.1 ICMP Reconnaissance Detection**

Reconnaissance is the initial phase of most cyberattacks. During this phase, attackers gather information about target systems before attempting exploitation.

One of the most common reconnaissance methods is ICMP-based host discovery. ICMP (Internet Control Message Protocol) is primarily used for network diagnostics and communication between network devices. Attackers often use ICMP Echo Requests, commonly known as ping requests, to determine whether a host is active.

When a system responds with an ICMP Echo Reply, it confirms that the host is reachable over the network. SOC analysts monitor ICMP traffic because repeated or abnormal ICMP requests may indicate reconnaissance activity.

During this project, ICMP packets exchanged between the attacker and victim systems were captured and analyzed.

#### **ICMP Traffic Capture in Wireshark:**

ICMP traffic analysis is an important part of network security monitoring because attackers often use ICMP packets during the reconnaissance phase of cyberattacks. By sending ICMP Echo Requests, attackers attempt to identify active hosts within a network and determine whether target systems are reachable. SOC analysts continuously monitor ICMP communication because abnormal ICMP activity may indicate network scanning, host discovery attempts, or suspicious communication behavior within the environment.



## SOC analysts often inspect ICMP packets to identify:

- Network reconnaissance attempts
- Host discovery activity
- Suspicious traffic frequency
- Unusual communication behavior

## 5.2 TCP SYN Port Scan Detection

After identifying active systems, attackers commonly perform port scanning to identify open services and communication ports.

A TCP SYN scan is one of the most widely used scanning techniques because it is relatively stealthy compared to full TCP connection scans. In a SYN scan, the attacker sends SYN packets to target ports and analyzes the response.

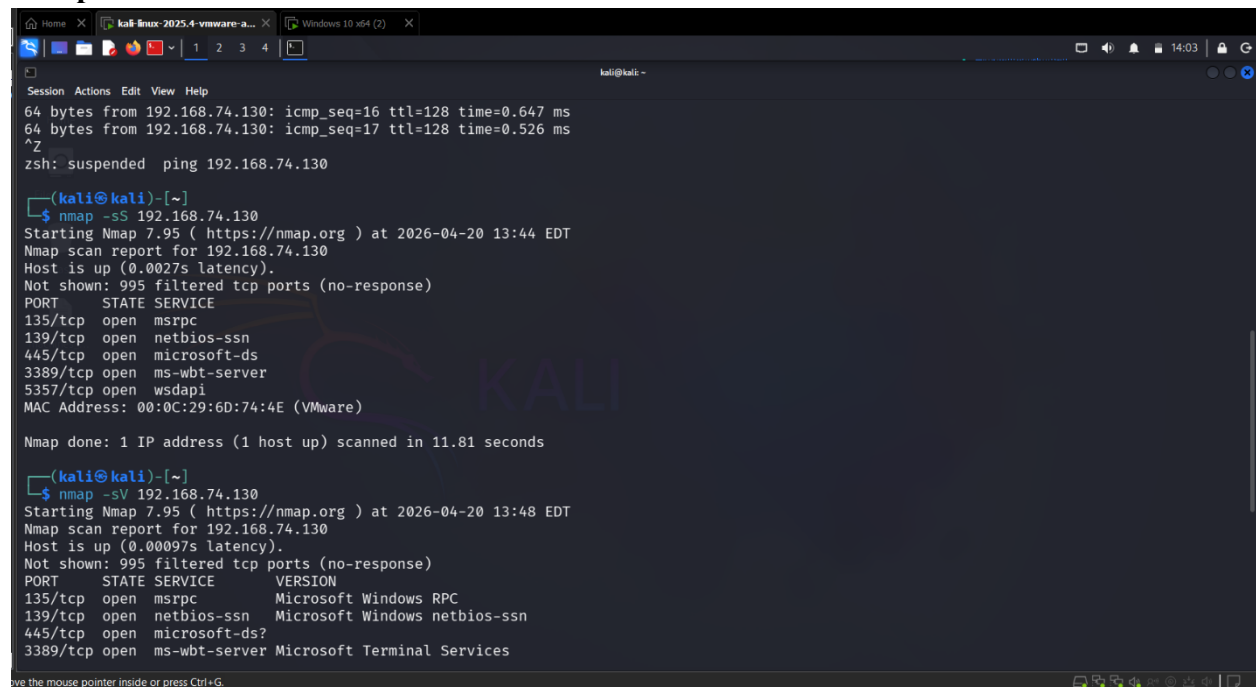
- SYN/ACK response indicates an open port
- RST response indicates a closed port

Nmap is a widely used network scanning tool capable of performing advanced reconnaissance and enumeration.

### Nmap SYN Scan Command

`nmap -sS 192.168.74.130`

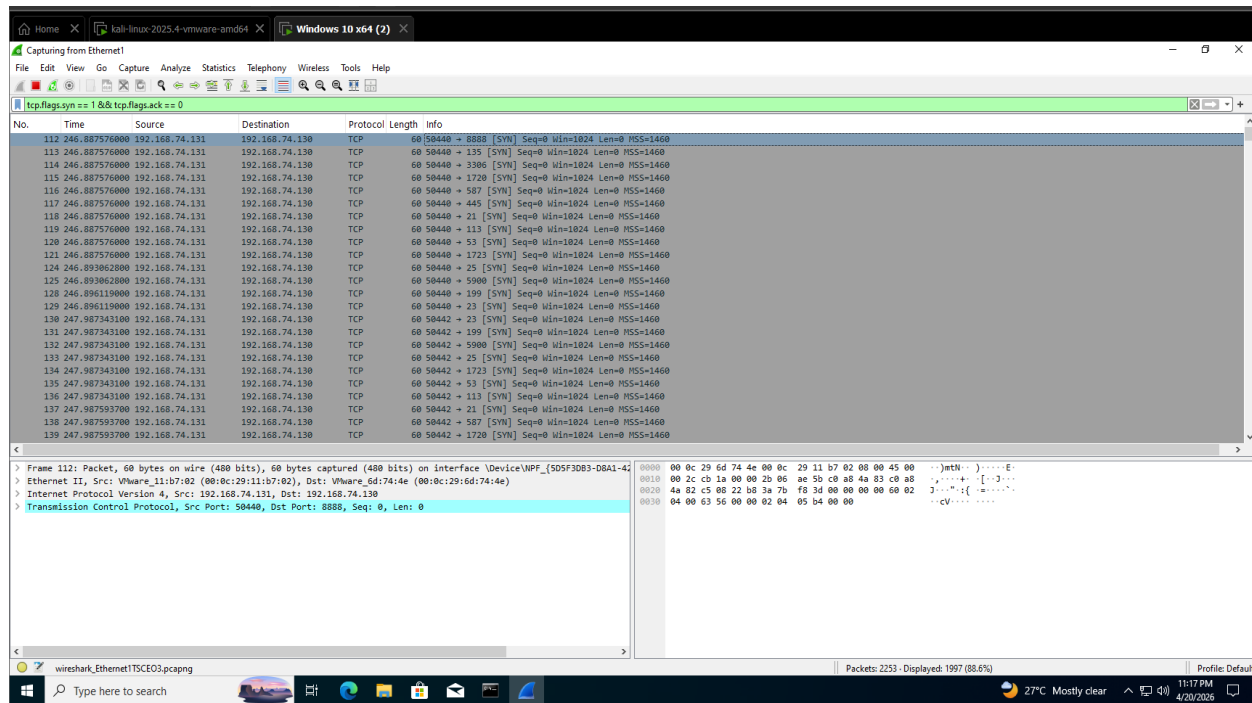
### Nmap SYN Scan Execution



```
kali@kali: ~  
$ nmap -sS 192.168.74.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-20 13:44 EDT  
Nmap scan report for 192.168.74.130  
Host is up (0.0027s latency).  
Not shown: 995 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
3389/tcp  open  ms-wbt-server  
5357/tcp  open  wsdapi  
MAC Address: 00:0C:29:6D:74:4E (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 11.81 seconds  
  
kali@kali: ~  
$ nmap -sV 192.168.74.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-20 13:48 EDT  
Nmap scan report for 192.168.74.130  
Host is up (0.00097s latency).  
Not shown: 995 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
```

The above figure shows the execution of a stealth SYN scan against the victim machine. The attacker attempted to identify accessible services running on the Windows 10 target system.

## SYN Packets Captured in Wireshark



The figure shows multiple TCP SYN packets generated during the Nmap scan.

### Wireshark Filter Used

tcp.flags.syn == 1 && tcp.flags.ack == 0

The packet capture demonstrates how SYN packets were transmitted sequentially to different ports on the victim machine.

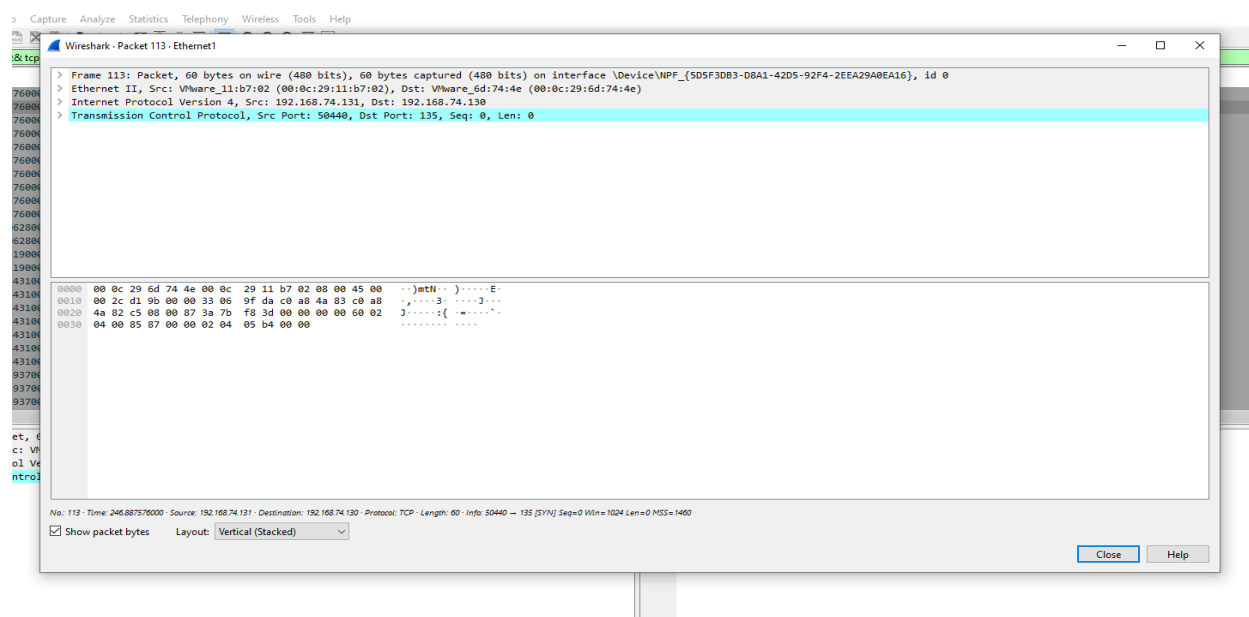
SOC analysts monitor SYN packet behavior because abnormal SYN activity often indicates:

- Port scanning
- Reconnaissance activity
- Worm propagation attempts
- Intrusion attempts

### TCP SYN Packet Detail Analysis

TCP SYN packet analysis plays a critical role in identifying reconnaissance and port scanning activities within enterprise networks. During a SYN scan, the attacker sends TCP SYN packets to multiple ports on the target machine to determine which services are active. The target system responds differently depending on the port status, allowing attackers to identify open, closed, or filtered ports without establishing a full TCP connection. Because of this stealthy behavior, SYN scans are widely used during the initial stages of cyberattacks.





The figure shows detailed analysis of TCP SYN packets including source port, destination port, flags, and sequence values.

Important TCP fields analyzed include:

- Source Port
- Destination Port
- TCP Flags
- Sequence Number
- Window Size
- Acknowledgment Number

The TCP flag analysis helps analysts understand connection behavior during scanning activities.

### 5.3 Service Enumeration Detection

Once open ports are identified, attackers attempt to determine the exact services and versions running on the target machine.

This process is known as service enumeration or service version detection.

Service enumeration is extremely important for attackers because it helps them identify:

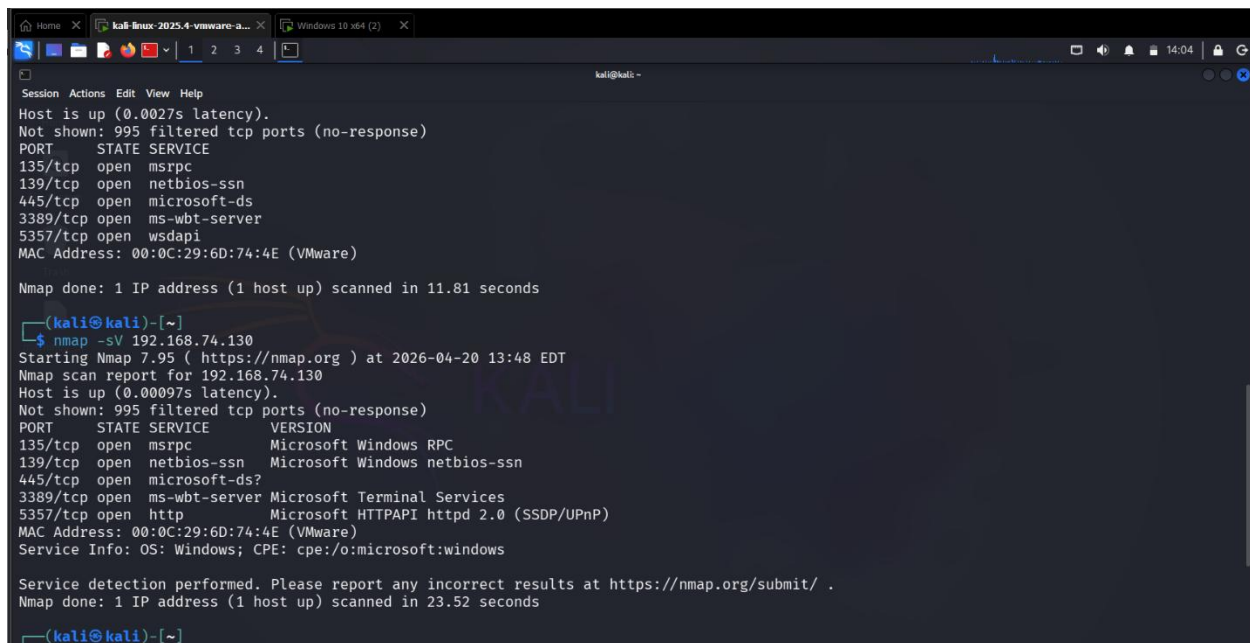
- Vulnerable software versions
- Misconfigured services
- Exposed administrative services
- Potential exploitation targets

Nmap service version detection was performed using the following command.

#### Nmap Service Enumeration Command

**nmap -sV 192.168.74.130**

#### Nmap Service Version Detection



```
Session Actions Edit View Help
Host is up (0.0027s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
5357/tcp   open  wsdapi
MAC Address: 00:0C:29:6D:74:4E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 11.81 seconds

(kali㉿kali)-[~]
$ nmap -sV 192.168.74.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-20 13:48 EDT
Nmap scan report for 192.168.74.130
Host is up (0.00097s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server Microsoft Terminal Services
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 00:0C:29:6D:74:4E (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.52 seconds

(kali㉿kali)-[~]
```

The figure shows service version detection results obtained from the victim machine.

## Detected Services

Port Service

135 MSRPC

139 NetBIOS

445 SMB

3389 RDP

5357 HTTPAPI

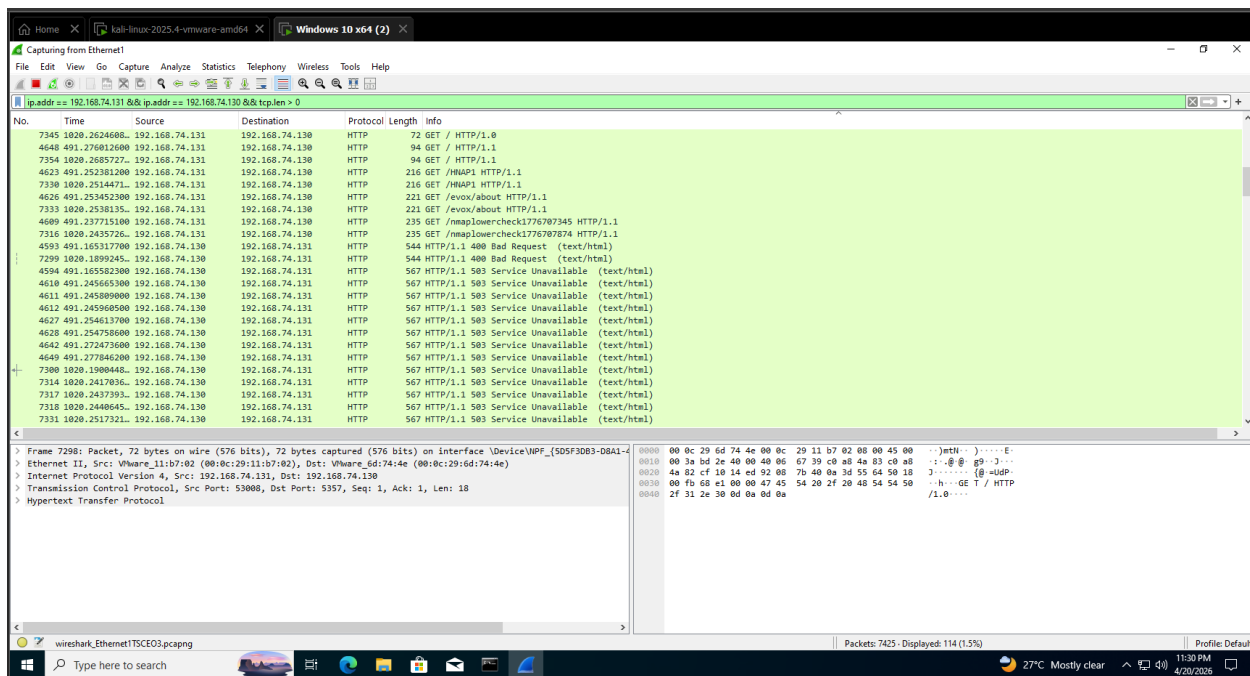
The identification of these services provides attackers with valuable information regarding the target environment.

For example:

- SMB services are frequently targeted during ransomware attacks
- RDP services are commonly targeted for remote access attacks
- RPC services may expose administrative functions

## HTTP Traffic Captured During Enumeration

HTTP traffic generated during service enumeration can provide valuable information about the target system and its active services. During the scan, HTTP requests and responses were captured in Wireshark, demonstrating how attackers gather information about web-based services running on the victim machine. Monitoring HTTP traffic helps SOC analysts identify suspicious enumeration attempts and unusual communication patterns within the network.



The figure shows HTTP requests generated during service enumeration activities.

HTTP traffic analysis allows SOC analysts to identify:

- Web-based reconnaissance
- Enumeration requests
- Suspicious HTTP headers
- Potential malicious communication

## 5.4 RDP Password Attack Detection

Remote Desktop Protocol (RDP) is a Microsoft protocol used for remote administration and remote desktop access.

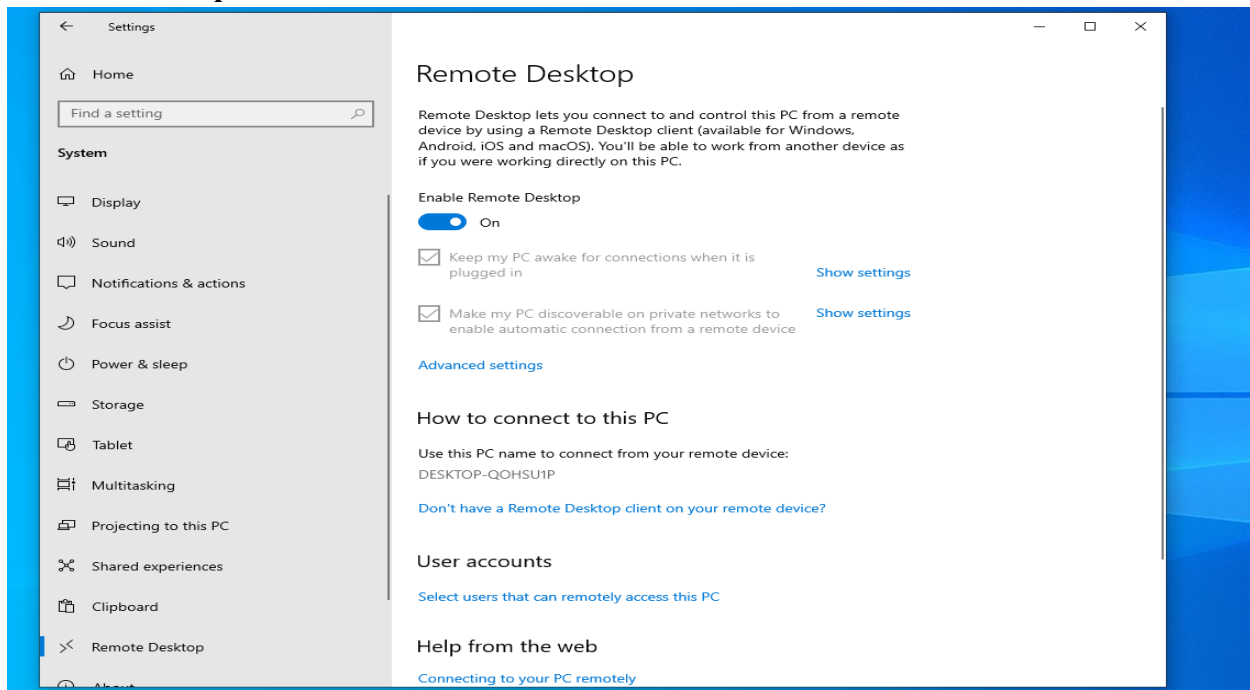
RDP is commonly targeted by attackers because exposed RDP services can allow unauthorized remote access to systems.

Attackers frequently use password attacks such as:

- Brute-force attacks
- Password spraying
- Credential stuffing

In this project, Hydra was used to simulate password attack attempts against the RDP service. RDP password attacks are commonly used by attackers to gain unauthorized remote access to systems exposed over the network. In this project, repeated authentication attempts against the Remote Desktop Protocol (RDP) service were simulated and analyzed using Wireshark.

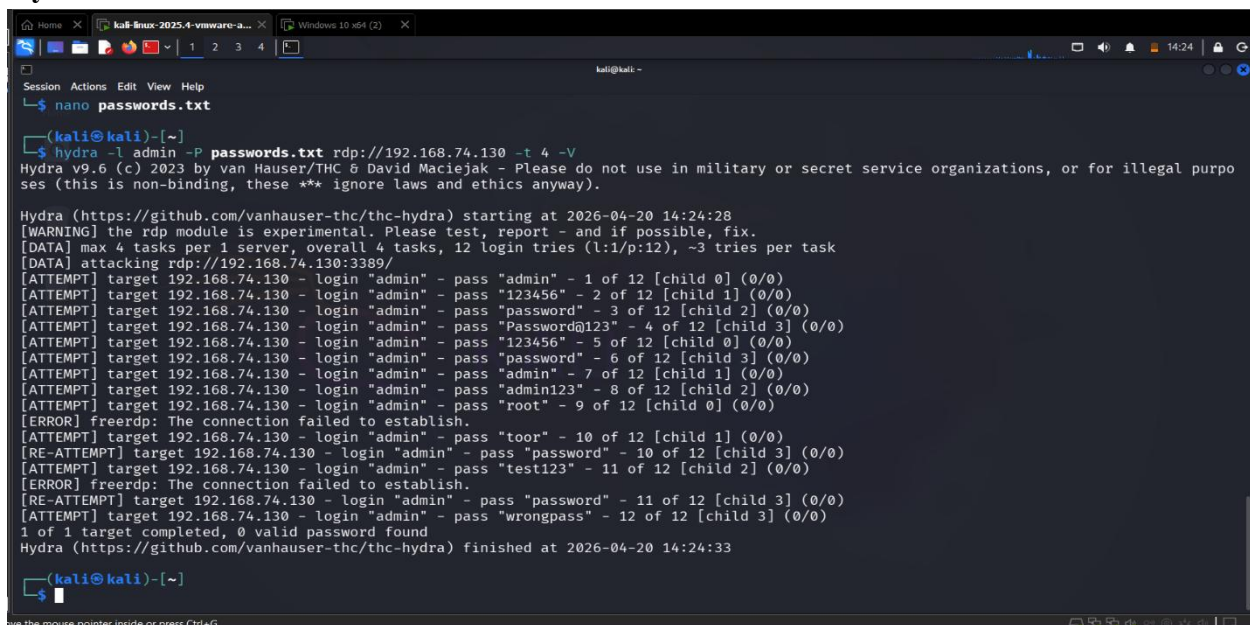
## Remote Desktop Enabled on Windows



The figure shows that Remote Desktop Protocol (RDP) was enabled on the Windows 10 victim machine.

Enabling RDP allowed the attacker machine to initiate authentication attempts against the target system.

## Hydra RDP Password Attack Execution



The figure shows the execution of Hydra password attack attempts against the RDP service running on the victim machine.

## Hydra Command Used:

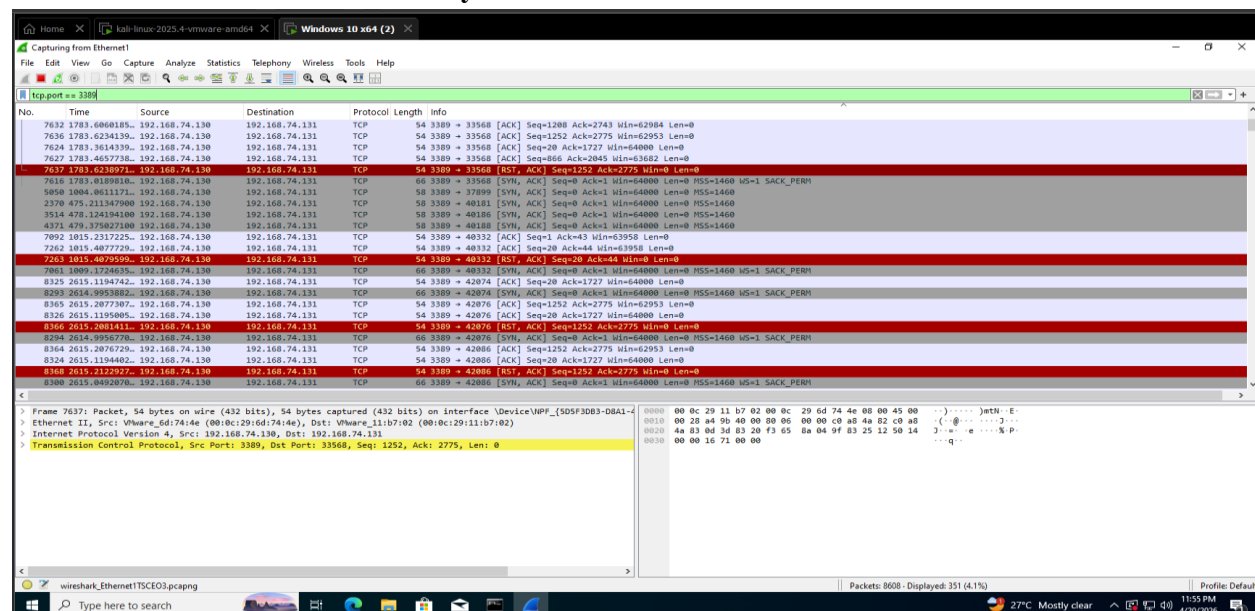
`hydra -l administrator -P rockyou.txt rdp://192.168.74.130`

Hydra attempted multiple password combinations against the RDP service.

SOC analysts monitor repeated authentication attempts because they often indicate:

- Unauthorized access attempts
- Password attacks
- Credential abuse
- Suspicious remote access activity

## RDP Traffic and TCP Reset Analysis in Wireshark



The figure shows RDP traffic and TCP Reset packets generated during failed authentication attempts.

## Wireshark Filter Used

`tcp.port == 3389`

Repeated RDP connection attempts generated multiple TCP communication events between the attacker and victim systems.

TCP Reset (RST) packets indicate terminated or rejected connections. During failed login attempts, RST packets may be generated due to unsuccessful session establishment.

SOC analysts monitor abnormal RDP traffic because exposed RDP services are frequently targeted in real-world attacks.

Common risks associated with exposed RDP services include:

- Unauthorized remote access
- Credential theft
- Ransomware deployment
- Lateral movement
- Persistence establishment

## **Traffic Analysis**

The captured traffic was analyzed to identify suspicious behavior associated with reconnaissance and authentication attacks.

### **Analysis Summary**

| Activity              | Protocol | Observation                                  |
|-----------------------|----------|----------------------------------------------|
| ICMP Reconnaissance   | ICMP     | Echo Request and Reply packets observed      |
| SYN Scan              | TCP      | Multiple SYN packets targeting several ports |
| Service Enumeration   | TCP/HTTP | Service identification requests observed     |
| RDP Attack            | TCP 3389 | Repeated connection attempts detected        |
| Failed Authentication | TCP      | RST/ACK packets generated                    |

---

### **7. Indicators of Compromise (IOCs)**

| IOC Type             | Value                |
|----------------------|----------------------|
| Attacker IP Address  | 192.168.74.131       |
| Victim IP Address    | 192.168.74.130       |
| Target Port          | 3389                 |
| Open Ports           | 135, 139, 445, 3389  |
| Reconnaissance Tool  | Nmap                 |
| Password Attack Tool | Hydra                |
| Protocols Observed   | ICMP, TCP, HTTP, RDP |

## **Mitigation Techniques**

The following mitigation techniques can help organizations defend against similar attacks:

- Restrict unnecessary open ports
- Disable unused network services
- Use strong passwords and account lockout policies
- Restrict RDP access using firewall rules
- Implement Intrusion Detection Systems (IDS)
- Monitor suspicious SYN scan activity
- Enable network logging and continuous monitoring
- Use VPN access instead of exposing RDP directly
- Regularly update and patch operating systems

## **Conclusion**

This project successfully demonstrated the detection and analysis of network attacks within a virtual SOC environment using Wireshark.

The project included reconnaissance detection, SYN scan analysis, service enumeration, and RDP password attack simulation. Network packets generated during these attacks were captured and analyzed successfully.

The analysis showed how SOC analysts can identify suspicious activities through packet-level monitoring and protocol inspection. The project also demonstrated the importance of early detection and network traffic analysis in cybersecurity operations.

This work provided practical experience in:

- Network traffic analysis
- Packet inspection
- Attack detection
- SOC investigation workflow
- Protocol analysis
- Cybersecurity monitoring techniques

## **References**

1. Wireshark Official Documentation
2. Nmap Official Documentation
3. Hydra Tool Documentation
4. TCP/IP Networking Concepts
5. VMware Workstation Documentation
6. Microsoft RDP Documentation

## **Appendix**

### **Wireshark Filters Used**

icmp

tcp.flags.syn == 1 && tcp.flags.ack == 0

tcp.port == 3389

## **Commands Used**

### **Nmap SYN Scan**

nmap -sS 192.168.74.130

### **Nmap Service Enumeration**

nmap -sV 192.168.74.130

### **Hydra RDP Attack**

hydra -l administrator -P rockyou.txt rdp://192.168.74.130